Pediatric Physical Assessment

Physical examination

Third Stage

Dr. Moustafa Ali Alshammari

Pediatric Nursing

Pediatric Physical Assessment

Collecting Data

- By observation
- Interviewing the parent
- Interviewing the child
- Physical examination

Bio-graphic Demographic

- Name, age, health care provider
- Parents name age /siblings age
- Ethnicity / cultural practices
- Religion / religious practices
- Parent occupation
- Child occupation: adolescent

Past Medical History

- Allergies
- Childhood illness
- Trauma / hospitalizations
- Birth history
- Did baby go home with mom / special care nursery
- Genetics: anything in the family

Current Health Status

- Immunizations
- Any underlying illness / genetic condition
- What concerns do you have today?

Review of systems

- Ask questions about each system
- Measuring data: growth chart, head and chest circumference, BMI
- Nutrition: breast fed, formula, eating habits
- Growth and development: How does parent think child is doing? Six questions

Physical Assessment

- General appearance & behavior
  - Facial expression
  - Posture / movement
  - Hygiene
  - Behavior
  - Development: grossly fits guidelines for age

Vital Signs

- Temperature: rectal only when absolutely necessary
- Pulse: apical on all children under 1 year
- Respirations: infant uses abdominal muscles
- Blood pressure: admission base line
- Height and weight and head circumference for 2 years and younger

Physical Assessment

- Skin, hair nails
- Head, neck, lymph nodes: fontanelles
- Eyes, nose, throat…look at palate and teeth
- Chest: auscultate for breath sounds and adventitious sounds
- Breasts: tanner scale
- Heart: PMI, murmurs

Physical Assessment

- Abdomen
- Genitalia: tanner scale, discharge, testicles
- Anus: inspect for cracks or fissures
- Musculoskeletal: Ortaloni maneuver / Barlows
- Feet / legs / back / gait

Neurological

- Glasgow coma scale
- Observe their natural state: Play games with them, especially children under 5 year
- CNS grossly intact: II – XII

Newborn reflexes

- Rooting: disappears at 3-4 months
- Sucking: disappears at 10 to 12 months
- Palmar grasp: disappears at 3 to 4 months
- Plantar grasp: disappears at 8 to 10 months
- Tonic neck: disappears by 4 to 6 months
- Moro (startle): disappears by 3 months
- Babinski: disappears by 2 years
- Stepping reflex: disappears by 2 months

Infant Exam

- Examine on parent lap
- Leave diaper on
- Comfort measures such as pacifier or bottle.
- Talk softly
- Start with heart and lung sounds
- Ear and throat exam last

Toddler Exam

- Examine on parent lap if uncooperative
- Use play therapy
- Distract with stories
- Let toddler play with equipment / BP
- Call by name
- Praise frequently
- Quickly do exam

Preschool Exam

- Allow parent to be within eye contact
- Explain what you are doing
- Let them feel the equipment

School-age Child

- Allow the older child the choice of whether to have a parent present
- Teaching about nutrition and safety
- Ask if the child has any concerns or questions
- How are they doing in school?
- Do they have a group of friends they hand out with?
- What do they like to do in their free time?

School-age Exam

- Allow choice of having parent present
- Privacy and modesty.
- Explain procedures and equipment.
- Interact with child during exam.
- Be matter of fact about examining genital area.

Adolescent

- Ask about parent in the room
- Should have some private interview time: time to ask the difficult questions
- HEADSS: home life, education, alcohol, drugs, sexual activity / suicide
- Privacy issues

Vital Signs

- Choose your words carefully when explaining vital sign measurements to a young child. Avoid saying, for example, “I’m going to take your pulse now.” The child may think that are going to actually remove something from his or her body. A better phrase would be “I’m going to count how fast your heart beats.”

Temperature



Position for taking axillary temperature.

Whaley and Wong

Temperature

- Use of tympanic membrane is controversial.
- Oral temperature for children over 5 to 6 years.
- Rectal temperatures are contraindicated if the child has had anal surgery, diarrhea, or rectal irritation.
- Check with hospital policy.

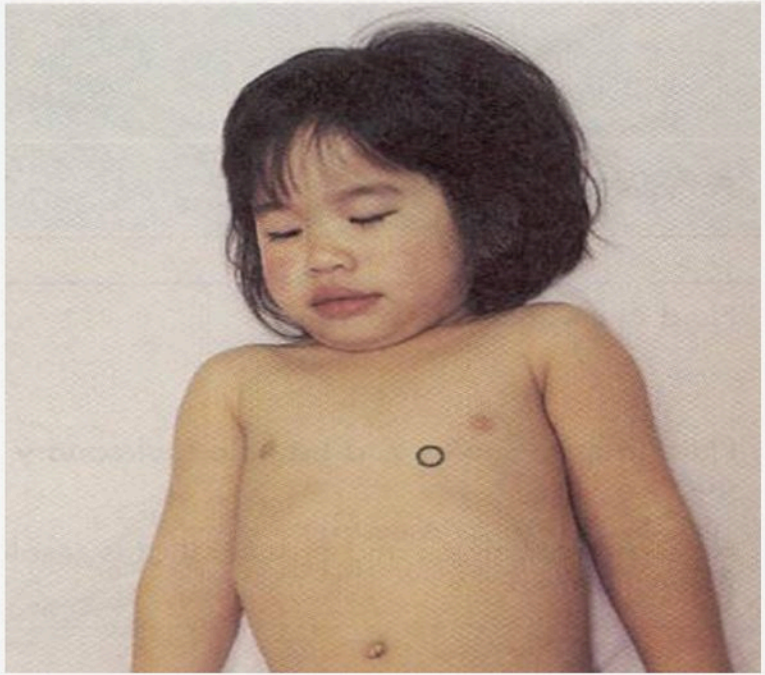
Pulse

- Apical pulse for infants and toddlers under 2 years
- Count for 1 full minute
- Will be increased with: crying, anxiety, fever, and pain

Pulse rates

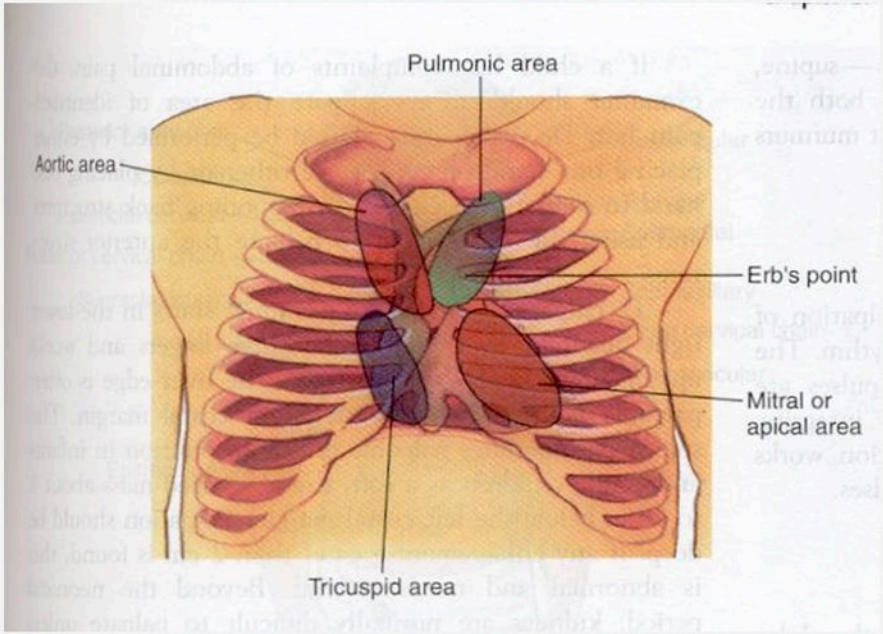
- Neonate: 70 – 190
- 1-year: 80 – 160
- 2-year: 80-130
- 4-year: 80 – 120
- 6-year: 75-115
- 10-year: 70-110
- 14-year: 65 – 105 / males 60 – 100
- 18-year: 55-95 / males 50 - 90

Apical Pulse



In child younger than 7 years.

Heart Sounds



See table 6-5, Bowden & Greenberg text

Auscultating Heart Sounds



Pillitteri

Respiratory

- Count for one full minute
- May want to do before you wake the infant up
- Rate will be elevated with crying / fever
  - Pre-term: 40 – 60
  - Newborn: 30 – 40
  - Toddler: 25
  - School-age: 20
  - Adolescent: 16

Panic levels: < 10 or > 60

Clinical Tip

- To accurately assess respirations in an infant or small child wait until the baby is sleeping or resting quietly.
- You might need to do this before you do more invasive exam.
- Count the number of breaths for an entire minute.

Blood Pressure

- The width of the rubber bladder should cover two thirds of the circumference of the arm, and the length should encircle 100% of the arm without overlap.
- Crying can cause inaccurate blood pressure reading.
- Consider norms for age.

Blood Pressure Cuff

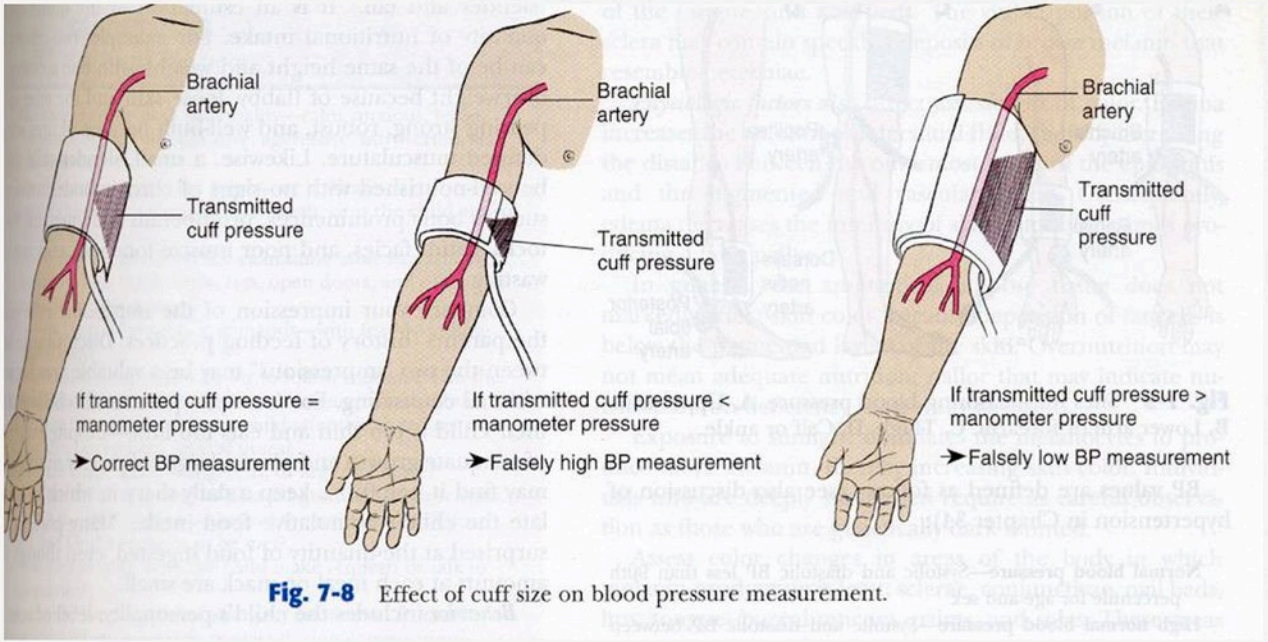


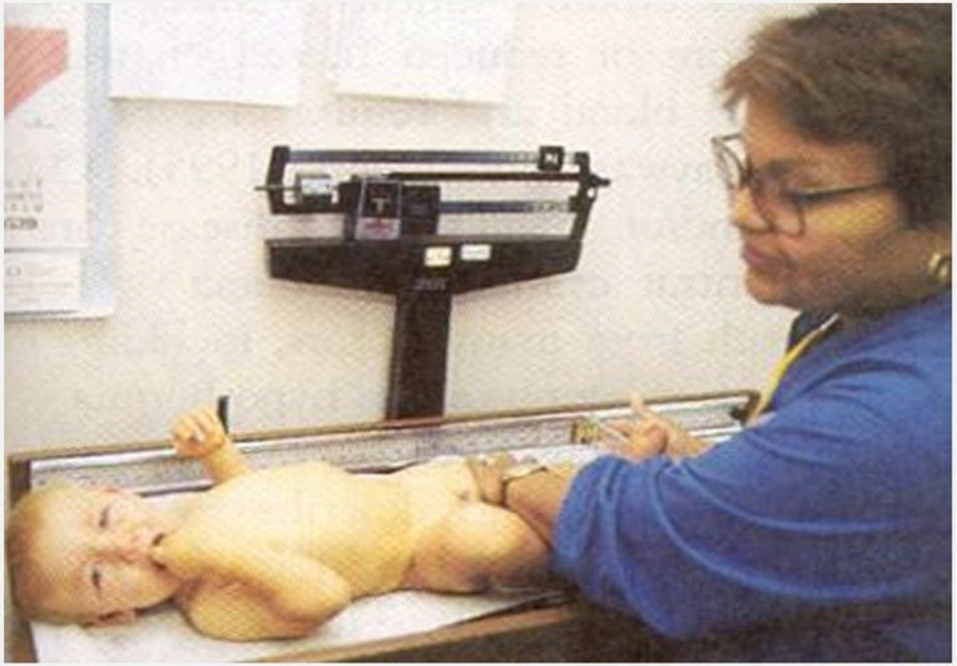
Fig. 7-8 Effect of cuff size on blood pressure measurement.

Whaley and Wong

Height

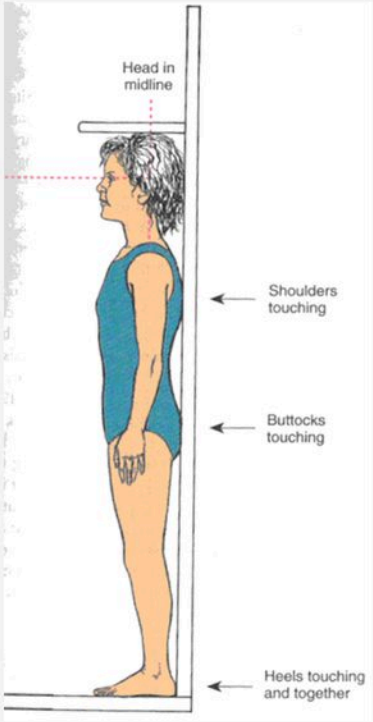
- Needs to be recorded on a growth chart
- Gain about an inch per month
- Deviation of height on either extreme may be indication for further investigation: endocrine problems

Height Measurement



Infants head is against end point and legs fully extended.

Height Measurement



Child is measured while standing in stocking or bare feet with the heels back and shoulders touching the wall.

Weight



Note close proximity of nurses hands for safety

Weight

- Needs to be recorded on a growth chart
- Newborn may lose up to 10% of birth weight in 3-4 days.
- Gains about ½ to1 oz per day after that
- Too much or too little weight gain needs to be further investigated.
- Nutritional counseling

Weight norms

- Double birth weigh by 5-6 months
- Triple birth weight by 1 year

Nutrition

- How much formula?
- How often being breast fed?
- Solid foods: 4 to 6 months of age
- What are they eating?
- Over 1 year: How much milk vs solid foods

Nutrition

- School age: typical diet
- Favorite foods
- I always child if I were to ask their mom what do they need to eat more of what would she say?

Nutrition

- Most common nutritional problems:
  - Iron deficiency anemia
  - Obesity
  - Anorexia

Head Circumference



Head circumference is measured by wrapping the paper tape over the eyebrows and the around the occipital prominence.

Head

- Needs to be measured until age 2 years
- Plot on growth curve
- Check fontales:
  - Anterior: 12 to 18 months
  - Posterior: closes by 2-5 months
- Shape: flat headed babies due to back-to-back sleep position

Mouth

- Palate
- Condition of teeth
- Number of teeth
- No teeth eruption by 12 months think endocrine disorder
- Appliances
- Brushing / visit to dentist

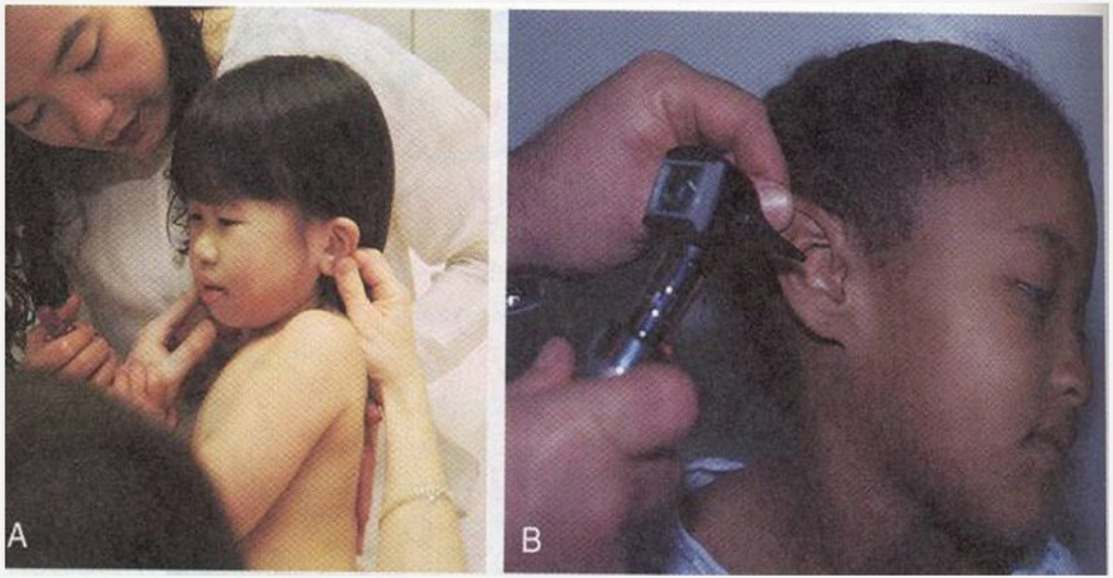
Eyes

- Check for red-reflex
- Can the infant see: by parent report
- Strabismus:
  - Alignment of eye important due to correlation with brain development
  - May need to corrected surgically
- 5-year-old and up can have vision screening
  - Refer to ophthalmologist if there are concerns

Common eye infections:

- Conjunctivitis:
  - A red-flag in the newborn may be STD from travel down the birth canal
  - Pre-school: number one reason they are sent home: wash with warm water / topical eye gtts
  - Inflammation of eye: history of juvenile arthritis

Ear Exam



Pinna is pulled down and back to straighten ear canal in children under 3 years.

Otitis Media

- Most common reason children come to the pediatrician or emergency room
- Fever or tugging at ear
- Often increases at night when they are sleeping
- History of cold or congestion

Otitis

- ROM: right otitis media
- LOM: left otitis media
- BOM: bilateral otitis media
- OME: Otitis media with effusion (effusion means fluid collection)

Pleural effusion, effusion of knee

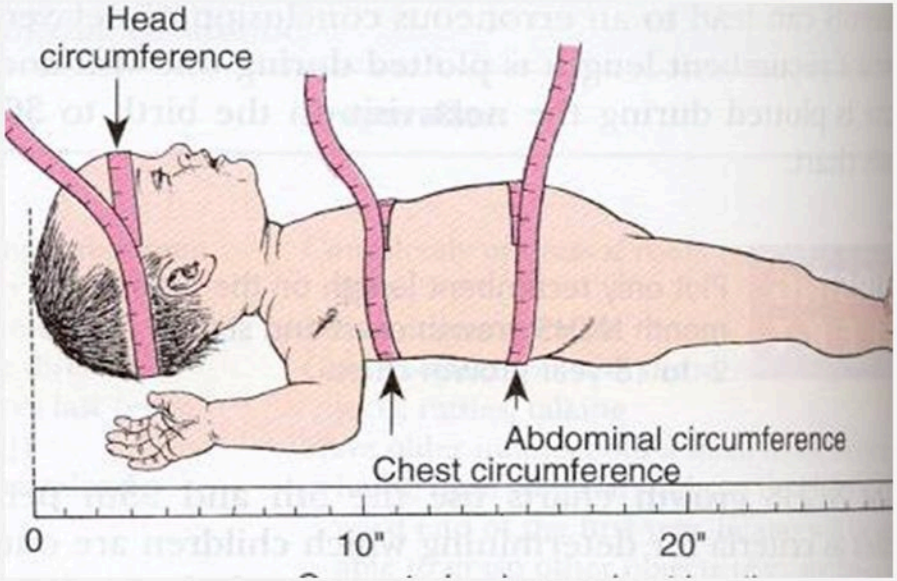
Why a problem?

- Infection can lead to rupture of ear drum
- Chronic effusion can lead to hearing loss
- OM is often a contributing factor in more serious infections: mastoiditis, cellulitis, meningitis, bacteremia
- Chronic ear effusion in the early years may lead to decreased hearing and speech problems

Management

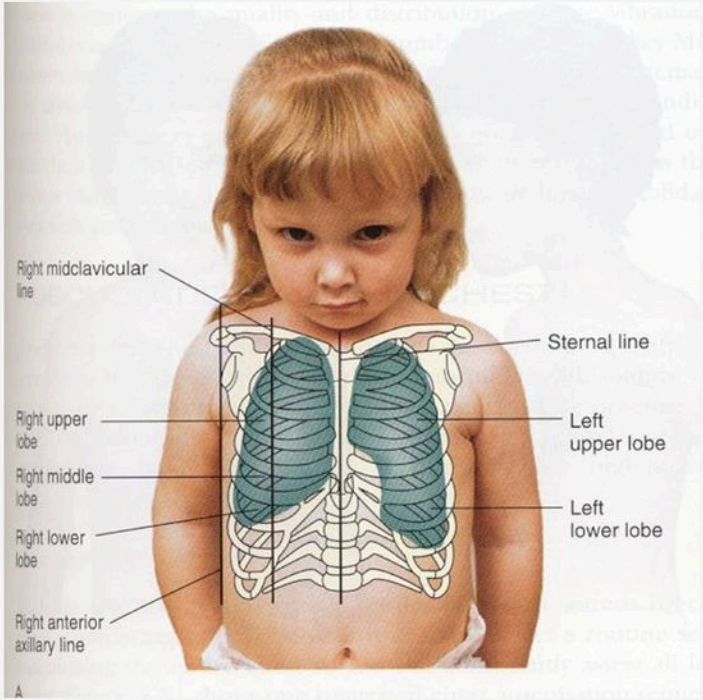
- Oral antibiotics: re-check in 10 days
- Tylenol for comfort
- Persistent effusion:
  - PET: pressure equalizing tubes
  - Outpatient procedure
  - Need to keep water out of ears
  - Hearing evaluation
  - Speech evaluation

Head, chest, and abdominal circumference.



Whaley and Wong

Child Chest



Ball and Bindler

Chest exam

- A high percentage of admissions to hospital are respiratory: croup, bronchitis, pneumonia, and asthma
- In the infant it is hard to separate upper air-way noises from lower air-way noises.
- How does the child look? Color, effort used to breathe

Possible Sites of Retractions

Observe while infant or child is quiet.



Bowden & Greenberg

Chest assessment

- Retractions
  - Subcostal
  - Intercostal
  - Sub-sternal
  - Supra-clavicular

Red flags: grunting / nasal flaring

Wheeze or Stridor

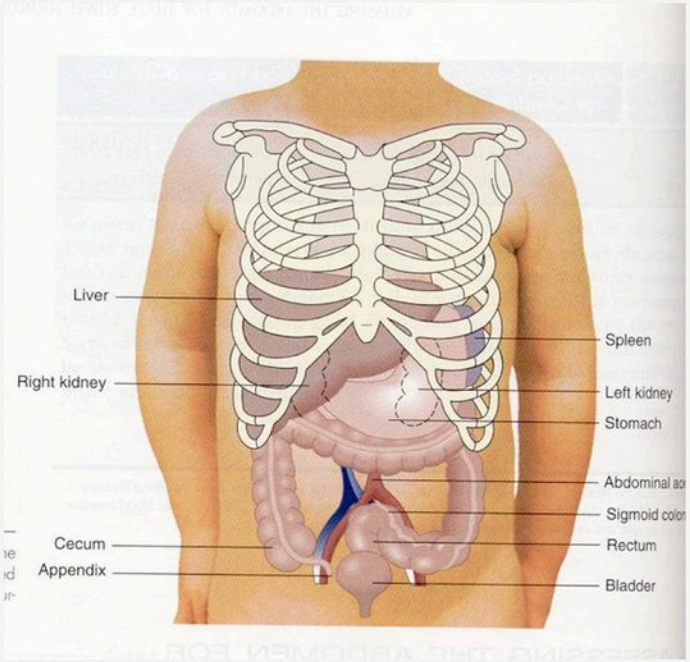
- Wheezes occur when air flows rapidly through bronchi that are narrowed nearly to the point of closure.
- Wheezes is lower airway
  - Asthma = expiratory wheezes
- A stridor is upper airway
  - Inflammation of upper airway or FB

Abdominal Girth



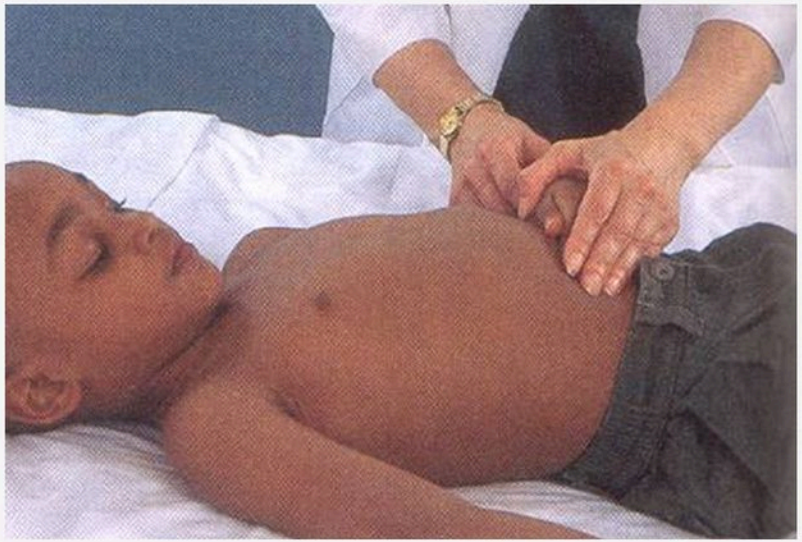
Abdominal girth should be measured over the umbilicus Whenever possible.

Abdomen



Ball & Bindler

Abdominal Assessment



Pillitteri

Clinical Tip

- Inspection and auscultation are performed before palpation and percussion because touching the abdomen may change the characteristics of the bowel sounds.

Bowel Sounds

- Normally occur every 10 to 30 seconds.
- Listen in each quadrant long enough to hear at least one bowel sound.
- Absence of bowel sounds may indicate peritonitis or a paralytic ileus.
- Hyperactive bowel sounds may indicate gastroenteritis or a bowel obstruction.